

PFMEA Failure Mode Library - Automotive Manufacturing

Common Failure Modes | Quality Head Agents

Machining Failures

Dimension out of tolerance | Surface finish not met | Burr not removed | Incorrect hole location | Thread defect | Flatness/roundness out of spec | Tool breakage not detected | Wrong material machined

Welding Failures

Incomplete weld / under-weld | Porosity / blow holes | Undercut | Cracks (hot or cold) | Wrong electrode grade | Burn-through | Incorrect weld sequence | Failed destructive test

Heat Treatment Failures

Under-hardness (below min spec) | Over-hardness (brittle) | Decarburization | Distortion beyond tolerance | Case depth out of spec | Quench cracking | Wrong cycle time

Assembly Failures

Missing component | Wrong component fitted | Under-torque / over-torque | Incorrect orientation | Adhesive not cured | Press force out of range | Damaged during assembly | Label missing or wrong

Inspection Failures

Out-of-calibration gauge used | Skip lot not followed | Wrong revision drawing used | Attribute check missed | SPC out of control not actioned | Measurement error | Wrong sample size

Packaging Failures

Incorrect label (wrong P/N, lot, qty) | Packaging damage in transit | Mixed parts in container | Wrong pack quantity | VCI film missing for rust protection | Incorrect container type for customer